PYTHON ACTIVITY BASED LEARNING

PROBLEM-5:

Sports Tournament Ranking System: Store match results and team scores, calculate total points and tie-break values, and produce rankings under a specified competition policy.

Objective:

To develop a Sports Tournament Ranking System that stores match results and team scores, calculates total points and tie-break values, and ranks teams according to predefined competition rules to determine the tournament standings accurately and fairly.

Algorithm

1. Start.

2.Read the number of teams.

3.Input all team names.

4.Read the number of matches.

5.For each match, input both team names and their scores.

6.Initialize each team's points, goals scored (GF), and goals conceded (GA) to zero.

7.Process every match:

- Update goals scored and goals conceded for both teams.
- Award 3 points to the winning team.
- Award 1 point to each team in a draw.

8. Calculate the goal difference for each team using \(GD = GF - GA\).

9.Sort the teams in descending order based on:

- Total points.
- Goal difference.
- Goals scored.

10.Calculate the number of matches played by each team.

11. Display the final rankings with rank, team name, matches played, points, GF, GA, and GD.

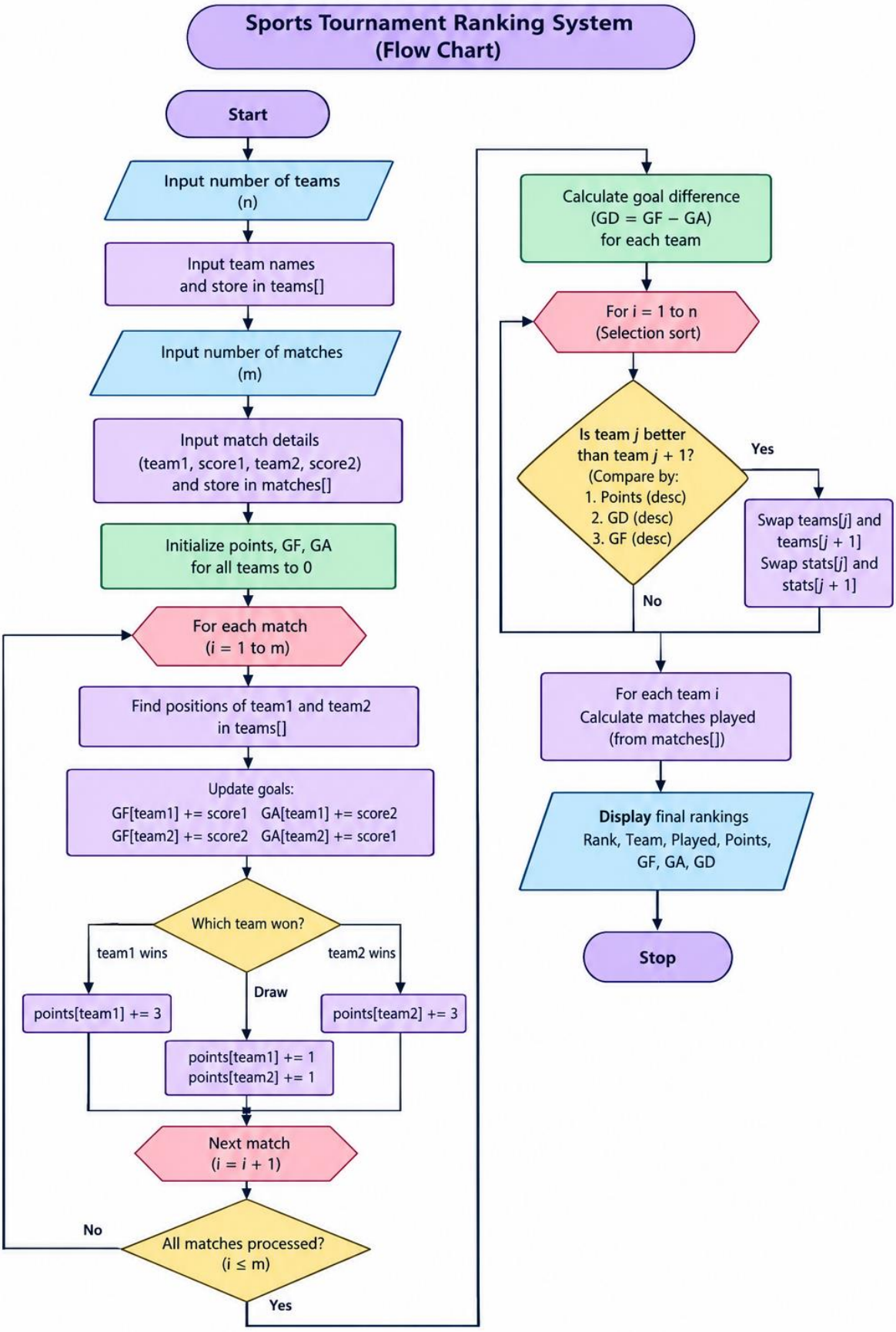
12. Stop.

Psudocode:

```
BEGIN
INPUT number_of_teams
FOR i = 1 TO number_of_teams
    INPUT team_name
    STORE team_name
END FOR
INPUT number_of_matches
FOR each match
    INPUT team1, score1, team2, score2
    STORE match details
END FOR
INITIALIZE points, GF, GA to 0 for all teams
FOR each match
    FIND positions of team1 and team2
    GF[team1] = GF[team1] + score1
    GA[team1] = GA[team1] + score2
    GF[team2] = GF[team2] + score2
    GA[team2] = GA[team2] + score1
    IF score1 > score2 THEN
        points[team1] = points[team1] + 3
    ELSE IF score2 > score1 THEN
        points[team2] = points[team2] + 3
```

```
    ELSE
        points[team1] = points[team1] + 1
        points[team2] = points[team2] + 1
    END IF
END FOR
FOR each team
    GD = GF - GA
END FOR
SORT teams by:
    1. Points in descending order
    2. Goal difference in descending order
    3. Goals scored in descending order
FOR each team in sorted order
    CALCULATE matches played
    DISPLAY rank, team name, matches played,
            points, GF, GA, GD
END FOR
STOP
```

Flowchart:


Sports Tournament Ranking System
(Flow Chart)
Start
Input number of teams
(n)
Input team names
and store in teams[]
Input number of matches
(m)
Input match details
(team1, score1, team2, score2)
and store in matches[]
Initialize points, GF, GA
for all teams to 0
For each match
(i = 1 to m)
Find positions of team1 and team2
in teams[]
Update goals:
GF[team1] += score1 GA[team1] += score2
GF[team2] += score2 GA[team2] += score1
Which team won?
team1 wins
Draw
team2 wins
points[team1] += 3
points[team1] += 1
points[team2] += 1
points[team2] += 3
Next match
(i = i + 1)
All matches processed?
(i ≤ m)
No
Yes
Calculate goal difference
(GD = GF − GA)
for each team
For i = 1 to n
(Selection sort)
Is team j better
than team j + 1?
(Compare by:
1. Points (desc)
2. GD (desc)
3. GF (desc)
Yes
No
Swap teams[j] and
teams[j + 1]
Swap stats[j] and
stats[j + 1]
For each team i
Calculate matches played
(from matches[])
Display final rankings
Rank, Team, Played, Points,
GF, GA, GD
Stop

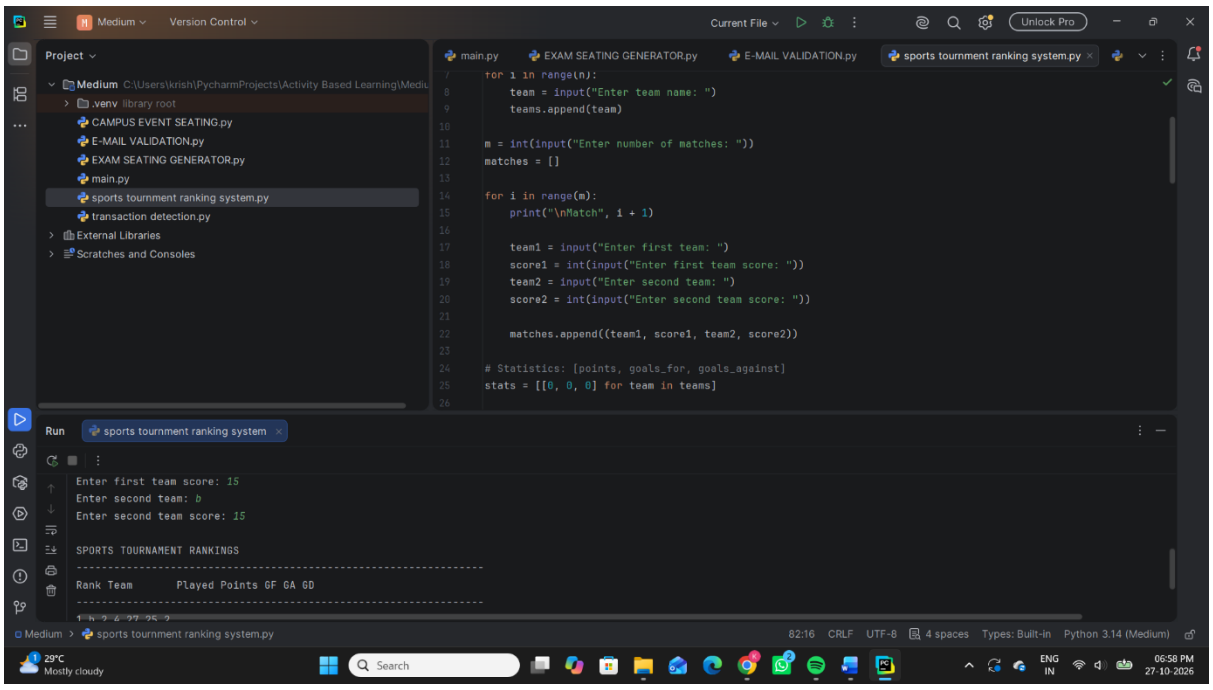
Python Program:

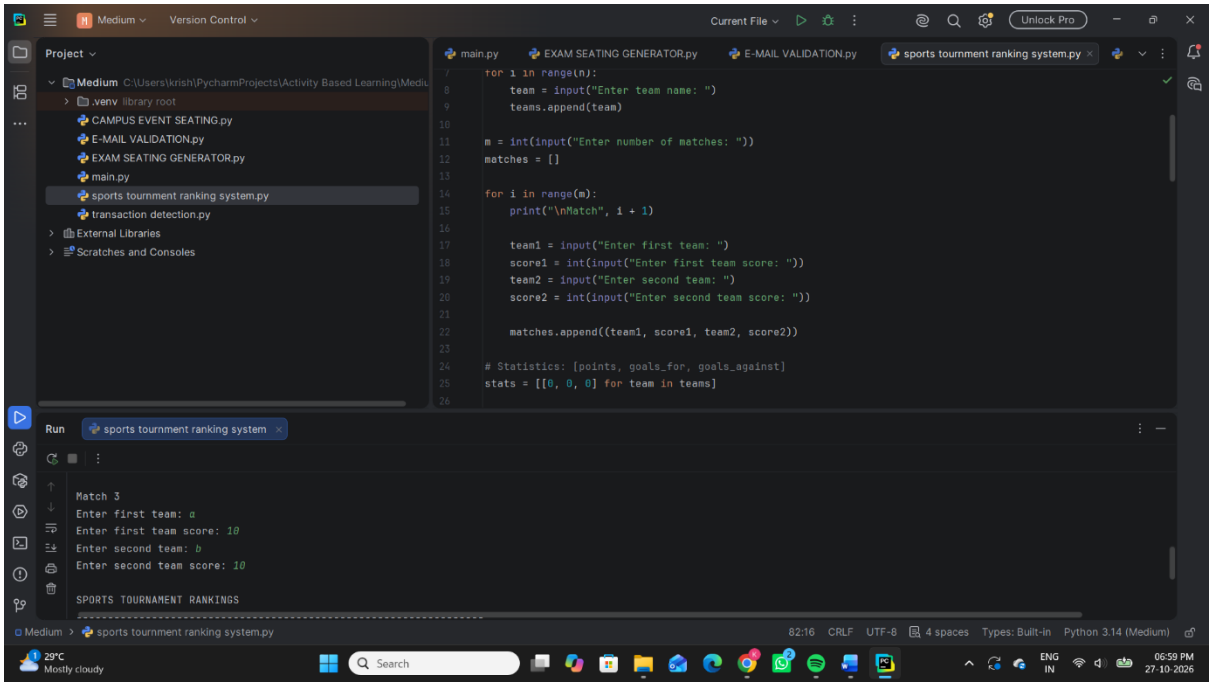
```
# Sports Tournament Ranking System
n = int(input("Enter number of teams: "))
teams = []
for i in range(n):
    team = input("Enter team name: ")
    teams.append(team)
m = int(input("Enter number of matches: "))
matches = []
for i in range(m):
    print("\nMatch", i + 1)
    team1 = input("Enter first team: ")
    score1 = int(input("Enter first team score: "))
    team2 = input("Enter second team: ")
    score2 = int(input("Enter second team score: "))
    matches.append((team1, score1, team2, score2))
# Statistics: [points, goals_for, goals_against]
stats = [[0, 0, 0] for team in teams]
# Process match results
for match in matches:
    team1, score1, team2, score2 = match
    index1 = teams.index(team1)
    index2 = teams.index(team2)
    stats[index1][1] += score1
    stats[index1][2] += score2
    stats[index2][1] += score2
```

```
    stats[index2][2] += score1
    # Award points
    if score1 > score2:
        stats[index1][0] += 3
    elif score2 > score1:
        stats[index2][0] += 3
    else:
        stats[index1][0] += 1
        stats[index2][0] += 1
# Sort by points, goal difference, and goals scored
for i in range(n):
    for j in range(n - i - 1):
        gd1 = stats[j][1] - stats[j][2]
        gd2 = stats[j + 1][1] - stats[j + 1][2]
        if (stats[j][0] < stats[j + 1][0] or
            (stats[j][0] == stats[j + 1][0] and gd1 < gd2) or
            (stats[j][0] == stats[j + 1][0] and
             gd1 == gd2 and
             stats[j][1] < stats[j + 1][1])):
            teams[j], teams[j + 1] = teams[j + 1], teams[j]
            stats[j], stats[j + 1] = stats[j + 1], stats[j]
# Display rankings
print("\nSPORTS TOURNAMENT RANKINGS")
print("-" * 65)
print("Rank Team       Played Points GF GA GD")
print("-" * 65)
```

```
for i in range(n):
    played = sum(
        1 for match in matches
        if match[0] == teams[i] or match[2] == teams[i]
    )
    points = stats[i][0]
    gf = stats[i][1]
    ga = stats[i][2]
    gd = gf - ga
    print(i + 1, teams[i], played, points, gf, ga, gd)
print("-" * 65)
```

Output Screen Shots:




Medium
Version Control
Current File
Unlock Pro
Project
Medium C:\Users\krish\PycharmProjects\Activity Based Learning\Mediu
.venv library root
CAMPUS EVENT SEATING.py
E-MAIL VALIDATION.py
EXAM SEATING GENERATOR.py
main.py
sports tournment ranking system.py
transaction detection.py
External Libraries
Scratches and Consoles
main.py
EXAM SEATING GENERATOR.py
E-MAIL VALIDATION.py
sports tournment ranking system.py
for i in range(n):
team = input("Enter team name: ")
teams.append(team)
m = int(input("Enter number of matches: "))
matches = []
for i in range(m):
print("\nMatch", i + 1)
team1 = input("Enter first team: ")
score1 = int(input("Enter first team score: "))
team2 = input("Enter second team: ")
score2 = int(input("Enter second team score: "))
matches.append((team1, score1, team2, score2))
Statistics: [points, goals_for, goals_against]
stats = [[0, 0, 0] for team in teams]
Run
sports tournment ranking system
Match 3
Enter first team: a
Enter first team score: 10
Enter second team: b
Enter second team score: 10
SPORTS TOURNAMENT RANKINGS
Medium
sports tournment ranking system.py
82:16
CRLF
UTF-8
4 spaces
Types: Built-in
Python 3.14 (Medium)
29°C
Mostly cloudy
Search
ENG
IN
06:59 PM
27-10-2026